

# Conclusion, Limits & Viva Pack

What was achieved, everything that is wrong with it, what novelty is missing, and the questions a jury will actually ask.

SHIPPED RMSE  
0.9006 degC

PS AUDIT  
16 pass / 0 fail / 1 blocked

TESTS  
925 passing, 9 skipped

WORKING SURFACES  
17, one port each

A jury does not test whether a team can present a good number. It tests whether a team knows what is wrong with its own work. This note is written so that every weakness in OceanEmbed is something you say first, in your own words, rather than something a reviewer discovers.

## 1. Conclusion - what was actually achieved

OceanEmbed reconstructs 15-level subsurface ocean temperature from 0 to 1000 metres across the North Indian Ocean at 0.25 degrees, daily, **from satellite surface observations alone**. The shipped model is TS-Cast-NIO stage 1: a 3-D CNN satellite encoder feeding a compact head, 548,582 parameters, trained on 388 consecutive days of seven-channel satellite input against a GLORYS12V1 target. The paper's climatology-prior decoder is built but **not shipped**: it was measured to cost accuracy, so the shipped network predicts the profile directly from the satellite latent.

| What was delivered                              | Evidence                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A working satellite-input reconstruction</b> | RMSE <b>0.9006 degC</b> , correlation <b>0.8809</b> , bias +0.1066 degC, skill <b>+0.2400</b> over climatology ( <b>+0.1494</b> where a real per-cell climatology exists), on <b>962 independent Argo profiles</b> and 12,736 depth comparisons under the seafloor_masked_v1 protocol. Beats climatology at <b>14 of 15</b> depths. |
| <b>Uncertainty on every value</b>               | A plus-or-minus 2 sigma band whose coverage is reported as a <b>range</b> (80.1% to 96.0% by depth) because the mean would hide an 80% depth at 50 m.                                                                                                                                                                               |
| <b>A complete system, not a model</b>           | <b>17 working surfaces</b> - validation, physics, events, 3-D cube, transect, acoustics, cyclone heat, case study, export, API, uncertainty, robustness and observation priority.                                                                                                                                                   |
| <b>Verification machinery</b>                   | <b>925 tests passing</b> , an <b>18-check freeze</b> on the shipped artifact, a <b>44-check</b> satellite-bundle verifier including a negative test that injects reanalysis and confirms it is caught, and a <b>17-row PS audit</b> that opens an artifact per clause.                                                              |
| <b>A measured account of its own limits</b>     | The training target's own error measured separately, so inherited error is distinguished from ours; a warm bias found twice by independent routes; and an unexplained regional penalty reported as unexplained.                                                                                                                     |

## 2. Scientific flaws in the shipped model

Ten of them, ranked by how much they matter, each with its magnitude and its owner [all VERIFIED]:

| #  | The flaw                                                | Magnitude                                                                 | Ours or inherited?                                                                                                                                                                                                             |
|----|---------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <b>The model runs warm</b>                              | bias +0.1066 degC overall, peaking at <b>+0.657 degC at 50 m</b>          | <b>Mostly inherited.</b> The GLORYS target is +0.1078 degC warm against the same floats and the model is -0.007 against its own target, so the average is the target's. The <b>+0.447 degC</b> the model adds at 50 m is ours. |
| 2  | The thermocline is the hardest depth                    | RMSE peaks at <b>1.22 degC at 100 m</b> ; correlation falls to 0.776      | <b>Mostly inherited</b> - the reanalysis itself scores 1.042 degC there - but <b>0.178 degC is ours</b> . The 0.023 degC an earlier draft quoted is a Phase-1 number and does not hold for this model.                         |
| 3  | The mixed layer (20-50 m) is worse than the reanalysis  | by +0.23 to +0.38 degC                                                    | <b>Ours</b> - and the one place effort would clearly pay.                                                                                                                                                                      |
| 4  | Climatology beats the model at 1000 m                   | by 0.055 degC                                                             | Ours, small, and labelled on the chart. 14 of 15 depths, not 15.                                                                                                                                                               |
| 5  | An Arabian Sea satellite penalty                        | +0.0341 degC, sign holds across 3 of 3 seeds                              | <b>Cause UNKNOWN after four tested hypotheses.</b> Reported as unexplained.                                                                                                                                                    |
| 6  | Uncertainty is improved, not calibrated                 | plus-or-minus 2 sigma covers <b>80.1% at 50 m</b> against a 95.4% nominal | Ours - mildly overconfident everywhere.                                                                                                                                                                                        |
| 7  | <b>The encoder has no missing-data channel</b>          | a gap and average water arrive at the network as the same number          | Ours, architectural, <b>unfixed</b> .                                                                                                                                                                                          |
| 8  | Stage 2 does not improve temperature on satellite input | mean +0.0042 inside a 0.0304 three-seed spread                            | Measured; not promoted, not frozen.                                                                                                                                                                                            |
| 9  | Stage-2 mixed layer and barrier layer are unusable      | MLD biased -14.12 m, barrier layer +8.88 m                                | Ours - caused by +0.19 psu of surface salinity bias against a 0.03 kg/m3 threshold.                                                                                                                                            |
| 10 | Stage-2 density calibration is unstable across seeds    | 1.245 / 1.281 / 1.491 - about 18% spread                                  | So no stage-2 uncertainty claim is made from a single run.                                                                                                                                                                     |

### 3. Data limits that do not move

These are not defects to be fixed with more effort. They are properties of the instruments and the grid.

- **Sub-mesoscale structure is unresolvable at 0.25 degrees** (about 25 km). A smooth field is not evidence that the ocean is smooth.
- **The satellite salinity product is blind to the Bay of Bengal freshwater plume.** It floors at 30.78 psu where the real signal reaches **6.43**. That is a sensor limit, not a modelling gap.
- **Nothing predicts salinity at depth from satellite inputs.** The seven channels carry a surface salinity only - which is why two structure fields are refused rather than approximated (Feature 5).
- **The SOFAR acoustic axis sits below our deepest level.** 94.75% of full-depth cells have their sound-speed minimum at 1000 m; the real axis in this basin is near 1500-2000 m. A basin-wide axis map would be a grid artifact 95% of the time, so it is refused (Feature 14).
- **The wind-stress product runs 2019-2022 only** and cannot reach the 2025-26 window, so the upwelling panel produces a signature and not an attribution.
- **Seasonal climatologies stay on the Phase-1 2019-2022 monthly record**, and say so. A four-year seasonal signal cannot be recomputed on a 388-day bundle without silently changing published magnitudes.
- **The data bundle is stale by about 74 days** and cannot be extended without real work - the pipeline requires a reanalysis target for every day and drops days lacking one, so every live day would be dropped and the bundle would come out empty.

### 4. What is NOT validated - say these before you are asked

- **Fronts are not validated.** The detector runs and its strongest July gradient lands where a front is expected, which is encouraging and is *not* evidence. No front climatology or published census was checked.
- **The Great Whirl identification is INFERRED.** That a 243 km August anticyclone holds station at 7.5 N, 53 E is VERIFIED in the data. That it *is* the Great Whirl rests on the standard description; no paper was re-read on this machine.
- **The Phase-1 out-of-distribution detector is not validated.** It flags 99.18% of the real test set because its reference file is a stale synthetic one. The detector is correct; the artifact is wrong.
- **The entire literature matrix is ABSTRACT-ONLY.** No methods section of any cited paper has been read on this machine. That is enough to position the work; it is **not** enough to quote a hyperparameter, claim a paper did not do something, or assert a numerical comparison.
- **Indian-language and regional literature has not been searched** - INCOIS, NIO Goa, IITM - and the sponsor knows that literature best.
- **Moored-buoy validation ran, and it is an IN-SAMPLE check.** The NOAA OSMC fetch succeeded on 2026-09-06 after earlier attempts timed out, and 46 series at 10 stations were scored - median RMSE 0.477 degC, median correlation 0.765. But those series span 2025-06 to 2026-06, which is the **training** period, and moored profiles feed the reanalysis the model is trained against. It shows the model tracks change at a fixed point; it is **not** independent validation and must not be quoted beside the 962-profile headline.

**A DISCREPANCY THAT WAS OPEN, NOW TRACED (2026-09-07)**

The Validation Lab headline (0.9638 degC over 879 profiles) and the freeze manifest (0.9006 degC over 962 profiles) are two different models on two different records: **src/phase2/validation/lab.py reads artifacts/argo\_error\_by\_depth.json**, the Phase-1 satellite-driven model scored against 2022 floats on the monthly grid, while the manifest records the v2 daily model against 2025-26 floats. Confirmed by reading the code, not inferred. If a juror asks, say exactly that.

## 5. What novelty is missing - the honest position

This section matters more than any other in a viva, because overclaiming novelty is the fastest way to lose a panel that knows the literature. Our own novelty matrix corrected its own initial assessment, and this is what it concluded.

| The idea                                                                | Verdict                                                                      | Prior art that already does it                                                                                                                         |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI reconstruction of subsurface temperature from surface data           | NOT NOVEL                                                                    | Meng et al. 2021 (JGR Oceans); DORS 2022 (Remote Sensing); FFPG-net 2025; TS-Cast 2026; NeSPReSO 2025.                                                 |
| Uncertainty-aware reconstruction                                        | ALREADY DONE - and better than ours                                          | TS-Cast 2026 predicts depth-dependent log error variances for temperature, salinity <i>and</i> density, parametric and calibrated.                     |
| Physics-guided reconstruction                                           | ALREADY DONE                                                                 | FFPG-net uses EOF modes; TS-Cast uses the equation of state.                                                                                           |
| Observation priority / where to measure next                            | ALREADY DONE - our seed matrix called this 'potentially novel' and was wrong | Objective-mapping optimisation of BGC Argo deployment (JTECH 40(11) 2023); optimal sensor placement via differentiable Gumbel-Softmax; FloatCast 2026. |
| 'First to reconstruct subsurface temperature in the North Indian Ocean' | NOT DEFENSIBLE                                                               | Global methods such as DORS 2022 already include this basin.                                                                                           |

**So what is missing?** A genuinely novel method. We did not invent a new architecture, a new uncertainty formulation or a new observation-targeting algorithm, and we do not pretend to have. Our observation-priority layer in particular is **a simple heuristic** - a geometric mean of two normalised factors - with no cost model, no float drift physics, no budget constraint, and one input (our sigma) that we ourselves measured to be mildly overconfident.

*The claim that survives: a North-Indian-Ocean-focused, independently-validated reconstruction system - surface to subsurface temperature with uncertainty, anomaly and an observation-priority layer - built and verified end to end. We are **not** proposing a new reconstruction method, and we do **not** claim novelty for uncertainty-guided observation targeting. Both are established fields whose state of the art exceeds our version.*

The exact wording to use. Do not strengthen it under pressure.

**Three things we added that the prior art above does NOT do.** Each is built on the frozen shipped model, each carries the control that decides whether its own headline is real, and each is measured [VERIFIED 2026-09-06]:

| Contribution                                | What is new                                                                                                                                                                                                                                                                                                       | Measured                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Static stability as a hard guarantee</b> | TS-Cast and everyone before it put density stratification in the <i>loss</i> . A soft penalty makes violations rare and cannot make them absent - which is why <b>no paper in this field reports a violation count</b> . We report ours, then remove them by projecting each profile onto the nearest stable one. | <b>805 of 13,468</b> adjacent pairs unstable (5.98%) in 597 of 962 columns; <b>12,153 of 165,648</b> (7.34%) across the basin. After projection: <b>0</b> , re-verified. Cost <b>+0.0002 degC</b> . A soft penalty trained for comparison cuts them 400-fold but <b>still does not reach zero</b> , and costs <b>+0.0220 degC</b> - 100x more. |
| <b>An observability field</b>               | Everyone reports where their model is inaccurate. Nobody separates <i>we are weak here</i> from <i>the surface carries no signal about this depth</i> . We differentiate the frozen model and show which satellite channel is doing the work at each depth.                                                       | The model reads <b>SST for the mixed layer and sea-surface height for the thermocline</b> (48% of the response at 100-125 m) - untaught, and physically correct, since SSH is the depth-integrated signal of thermocline displacement.                                                                                                         |
| <b>Latent-space assimilation</b>            | The literature treats Argo as the scoring rubric and never feeds it back. We optimise the 128-number latent - network frozen, no retraining - until the decoder reproduces a real float, then propagate by similarity in <i>latent</i> space rather than by distance.                                             | MAE <b>0.5441</b> -> <b>0.5291</b> at latent-similar cells, while random recipients get 0.0241 worse and the least-similar decile 0.0745 worse. Still <b>+0.0065 degC</b> at recipients 500 km away.                                                                                                                                           |

## TWO OF THE THREE CARRY A RESULT WE DID NOT WANT, AND THAT IS THE POINT

The observability work set out to show that our errors sit *below* the information floor - that deep error is the surface having nothing left to say. **It is refuted, at every testable depth, in the opposite direction:** low-sensitivity profiles have *smaller* errors, because low sensitivity marks quiet water close to climatology. And the assimilation gain is real but modest - about 2.8% - and mostly local. Both are on the page at the same size as the headline.

**What we can defensibly claim as contribution**, and it is worth saying clearly rather than apologetically:

- **Two measured disagreements with the paper we re-implemented**, on one seed each: its density constraint costs accuracy, and its FiLM decoder was measured to cost accuracy and is deliberately not shipped. A third - its 31-day input window - is **withdrawn**: that comparison predates our own leakage fix and has not been re-run.
- **The reanalysis ceiling measurement** - separating inherited error from ours - which nothing else in this project had ever done and which redirected the remaining effort.
- **The cloud-dropout finding**: a robustness experiment that instead measured a warm bias in the deliverable and explicitly rejected the flattering reading of its own curve.
- **The satellite-input discipline**: refusing to ship or quote a more accurate reanalysis-fed model, enforced by an assertion in the freeze script rather than by intention.
- **The system engineering**: 892 tests, an 18-check freeze, byte-identity manifests across two machines, and refusals encoded as behaviour rather than described in prose.

## 6. Engineering and process risks, live right now

- **Model checkpoints and data bundles are outside version control** (about 222 MB and 31 GB) and move between machines by zip. Checksum manifests are the only proof that the file on one machine is the one that was scored.
- **Two of the most flattering numbers in the project (0.8548 and 0.8593) have never been re-verified on this disk.** They are labelled as such - and they are comparators, not the deliverable.
- **Inference runs on CPU by default** - about 32 seconds per whole-field reconstruction - although a measured 4.2x GPU speedup exists. This is deliberate, so an exported file cannot differ in its last digits from the page beside it, but it is a real performance ceiling.
- **Seventeen ports is a lot of surface for a demo**, and nothing aggregates them into a single application.
- **The acceptance script carries one known pre-existing failure**, comparing two legacy Phase-1 artifacts with different profile-retention rules. Neither underwrites the shipped model, and the RMSE agreement it also checks passes at 0.0213 degC.

## 7. The corrections we made on ourselves

Several numbers in this project were believed, published internally, and wrong. Every one of them looked right. Mentioning this in a viva is not a confession - it is the strongest available evidence that the numbers still standing have been tested.

| What went wrong                                                                                      | How it was caught                                                                                                                                                                         | What changed                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A data leak.</b> Five of 304 training days read surface fields from the test period - 1.64%.      | The split assertion was correct; the input <i>window</i> was not, which is exactly why every test passed while it happened.                                                               | Every artifact from before the fix is marked <b>INVALID</b> and preserved verbatim rather than edited. Two independent fixes were written; the stricter one is canonical. |
| <b>A result reported from one seed.</b> A gradient loss appeared to improve the model by 0.019 degC. | Seed 42 improved under both weights while seeds 43 and 44 degraded. Run on one seed we would have reported an improvement and been wrong - <b>which had already happened once before.</b> | The sweep script now <b>refuses to declare anything</b> on fewer than three seeds.                                                                                        |
| <b>An SSH claim</b> based on 1 of 3 seeds, presented as 3 of 3.                                      | Internal review.                                                                                                                                                                          | Retracted, and the three-seed rule was made explicit.                                                                                                                     |
| <b>An empty float match explained as an empty ocean.</b>                                             | The float table simply held no rows for that year.                                                                                                                                        | The collocation engine now reports table coverage explicitly, so a gap in a table can never be read as a fact about the sea.                                              |
| <b>A false SOFAR detection</b> in about ten metres of water in the Palk Strait.                      | Clicking one cell on the page.                                                                                                                                                            | A full-column guard, plus a test that relaxes the guard on purpose and confirms the bad result returns.                                                                   |

## 8. Viva questions, with answers that hold up

### A. The headline result

#### Q. What exactly is your result, and against what?

**A.** RMSE 0.9006 degC, correlation 0.8809, bias +0.1066 degC, and +0.2400 skill against climatology - measured on 962 independent Argo profiles and 12,736 depth comparisons, using satellite inputs only. The profiles were never used in training. Where the cell has a real per-cell climatology - 895 of the 962 - the skill is +0.1494; the other 67 sit on a basin-mean fill that flatters any model.

#### Q. Is 0.9 degrees of error good?

**A.** It is meaningful in context and we can give you three. First, it is 24% better than climatology - 15% where the baseline is a real per-cell climatology - and it beats climatology at 14 of 15 depths. Second, most of the error at the hardest depth is the ceiling our training target sets: at 100 m the GLORYS reanalysis itself scores 1.042 degC against the same floats while we score 1.218, so 0.178 degC of that gap is ours and the rest is inherited. Third, it is achieved

from satellite surface data alone, where the reanalysis uses assimilated in-situ observations and supercomputing.

### **Q. Why is your headline not your best number?**

**A.** Our best number, 0.8548, is fed reanalysis inputs. The problem statement asks for satellite observations, so quoting it would present a reanalysis-fed model as satisfying a satellite requirement. The freeze script asserts the shipped artifact's input source reads 'satellite'. Against the reanalysis-fed stage-1 comparator the satellite model reads +0.026, +0.027 and -0.003 degC across three seeds - a mean cost of +0.017 whose sign does not hold, so by our own rule we do not claim a cost; it retains about 94% of the skill, and the two inputs are within seed noise.

### **Q. Where is the model worst?**

**A.** At 100 metres - the thermocline - where RMSE reaches 1.22 degC and correlation falls to 0.776. Most of that is inherited - the reanalysis scores 1.042 degC at the same depth - though 0.178 degC of it is ours. The error most clearly ours is in the mixed layer, 20 to 50 metres, where we are 0.23 to 0.38 degC worse than the reanalysis.

## **B. Method and model**

### **Q. Explain your architecture in one minute.**

**A.** Seven satellite channels over an 11-day window go into a 3-D convolutional encoder, which compresses them to a 128-number latent. That latent conditions a decoder whose *input* is the monthly climatology - so the network learns a correction to a physically sensible average rather than inventing a profile. Two heads come out: temperature at 15 depths, and a log-variance that becomes the uncertainty. 548,582 parameters in total.

### **Q. Why start from climatology instead of predicting directly?**

**A.** Because it is a far better-posed problem. Predicting a correction to a sensible average keeps the output physical where satellite evidence is thin, and it is the core idea of the paper we re-implemented. We kept it behind a flag so the claim stays testable.

### **Q. Why 11 days of input?**

**A.** Honestly: because an ablation chose it, and that ablation no longer stands. All three legs were run before we fixed a leakage bug in the sampler, the 31-day leg sits on our own do-not-quote list, and the two shorter legs' checkpoints were overwritten by the next run. So 11 days is what the shipped model uses and we cannot presently prove it is the best choice. Re-running the three windows under the embargo is a named open item, not a claim we are making today.

### **Q. How did you choose the encoder?**

**A.** A bake-off between four candidates - a centre-cell-only control, a 3-D CNN, a CNN with attention, and a vision transformer - all sharing an identical prediction head, so the encoder was the only variable. The 3-D CNN won.

### **Q. Your model is small. Is that not underfitting?**

**A.** It is deliberate and measured. At the paper's widths the model would be 7.1 million parameters - 13 times the capacity for our 100,000 samples - and it overfits by epoch 3 regardless of the loss function. We kept the paper's widths in the config for the record and ship the smaller model.

### **Q. Did adding physics to the loss function help?**

**A.** No, and we measured it on three seeds. At one weight it was clearly worse; at another the mean sat inside the seed noise and the sign did not hold. We then measured *why*: the model already reproduces 100.6% of the observed thermocline gradient, so the term was protecting structure that was never being lost. A null result, reported as one.

## **C. Data and inputs**

### **Q. How do we know your inputs are really satellite data?**

**A.** A bundle verifier runs 44 provenance and lineage checks, and it includes a negative test that deliberately injects reanalysis into all five satellite channels and confirms the check catches it. The PS audit also names each source

product by its catalogue identifier and its source type.

**Q. How much data did you train on?**

**A.** 388 consecutive daily steps from 1 June 2025 to 23 June 2026, with zero gaps, over a 100 by 240 grid - and a strict time-based train/test split with an embargo at the boundary.

**Q. What about missing data - clouds?**

**A.** We measured it as a designed experiment. Blanking sea-surface temperature the way cloud does, the error degrades monotonically past about 20% and reaches +0.50 degC at total blindness. The honest caveat is that our masking is random while real cloud is clustered and persistent, so a genuine overcast should hurt more. And the architectural gap is stated: the encoder has no way to know a pixel is missing.

**Q. Is your data current?**

**A.** No - the bundle ends 23 June 2026 and is about 74 days stale, and extending it is real work rather than a re-run, because the pipeline requires a reanalysis target for every day and would drop every live day. We would rather state that than imply an operational feed exists.

**D. Uncertainty**

**Q. Is your uncertainty calibrated?**

**A.** It is *improved*, not *calibrated*, and we use exactly those words. Our plus-or-minus 2 sigma band covers 91.2% of held-out observations against a nominal 95.4%. We report coverage as a range across depths - 80.1% to 95.5% - because the mean hides an 80% depth at 50 metres. We never label the band '95%', and the freeze check verifies that wording has not drifted.

**Q. How do you know the calibration is not fitted on the data it is scored on?**

**A.** The scaling is fitted on 3,423 profiles from the training window and reported on 908 profiles from the test window - disjoint in time. It was also fitted against the shipped checkpoint specifically, which the freeze script asserts.

**Q. Where is the model least certain?**

**A.** At 100 metres, median sigma 1.197 degC, and most certain at 500 metres at 0.265. The largest calibration correction is also at 100 metres. Two independent signals point at the same depth: the raw model is least sure at the thermocline and was most overconfident there.

**E. Novelty and prior art**

**Q. What is novel here?**

**A.** The *system*, not the method. We say that plainly. AI reconstruction of subsurface temperature is established prior art, uncertainty-aware reconstruction is done better by the paper we re-implemented, and observation targeting is a formally-optimised research area of which ours is a simplified heuristic. What we contribute is a region-focused, independently-validated, end-to-end verified system, three measured disagreements with the paper we re-implemented, and a measurement of our own training target's error that separates inherited error from ours.

**Q. You said nothing is novel, then you listed three contributions. Which is it?**

**A.** Both, and the distinction matters. The *reconstruction method* is prior art and we do not claim it. What we added afterwards is not: a hard static-stability guarantee where the field uses a soft penalty and consequently never reports a violation count; a per-depth, per-channel observability field showing the model reads sea-surface height for the thermocline untaught; and inference-time assimilation of a real float into the frozen latent. Two of those three came back with results we did not want, and we report those too.

**Q. Your stability projection - does it not just hide a bad model behind a filter?**

**A.** It cannot, and that is a property of the operator rather than a promise. Isotonic projection is exactly the identity on a profile that was already stable, so it cannot touch a good column. And it costs +0.0002 degC of RMSE against

independent floats, so it is not buying the guarantee with accuracy. What it does buy is that 5.98% of our adjacent level pairs - in 62% of columns - stop being physically impossible.

**Q. So you just re-implemented a paper?**

**A.** We re-implemented a published description - their code was never released and nothing is copied - and then measured where it does not hold at our data scale. Three of their choices cost us accuracy: the 31-day window, the density constraint, and the FiLM decoder. We built the FiLM decoder, measured it, and refused to ship it, with the refusal encoded as an error message in the code.

**Q. Have you read the papers you cite?**

**A.** The abstracts and landing pages, and we tag the whole literature matrix ABSTRACT-ONLY for exactly that reason. That is enough to position the work and not enough to quote a hyperparameter or assert a numerical comparison, and we do neither.

**Q. Does your observation-priority feature tell INCOIS where to deploy floats?**

**A.** No, and that boundary is enforced rather than remembered. The sanctioned wording lives in the module, is rendered verbatim on the page, and a test searches both files for any phrasing claiming to tell anyone where to deploy - and fails unless it is negated. No numerical test can catch a rhetorical error, so the rhetoric is tested too.

## **F. Verification and reproducibility**

**Q. How do we know the model on screen is the one you scored?**

**A.** Run the freeze check. It re-verifies eighteen properties of the shipped artifact rather than trusting the manifest, including the checkpoint's SHA-256 hash, the input source, and the claim wording used for uncertainty. There is also a byte-identity manifest that has been proven to fail on a tampered file.

**Q. How much of this is tested?**

**A.** 892 tests pass, 9 are skipped. More usefully: several tests are *falsification* tests - they relax a guard on purpose and confirm the bad behaviour returns, so a guard cannot be quietly removed. One test parses the source code and fails if a particular mask ever starts being returned, because that would make a page's published framing out of date.

**Q. Can we reproduce your numbers?**

**A.** Yes, and each note in this pack gives the exact command. The seed and configuration are saved with every run, the acceptance script opens artifacts rather than trusting them, and the re-scoring script reads a checkpoint from disk rather than trusting its sibling results file.

**Q. What is the biggest engineering risk you carry?**

**A.** That checkpoints and data bundles live outside version control - about 31 GB - and move between machines by zip. The checksum manifests are the only thing that proves the files on one machine are the ones that were scored.

## **G. Impact and deployment**

**Q. Who would actually use this?**

**A.** INCOIS and MoES as the sponsor; IMD and cyclone forecasters for the subsurface heat that drives intensification; fisheries for thermocline, front and upwelling information; the navy and coast guard for acoustics; and climate monitoring for ocean heat content. Each feature note names its own users.

**Q. What is the single most important application?**

**A.** Tropical Cyclone Heat Potential. It is what decides whether a storm intensifies, it has always required subsurface observations, and we produce it daily across the whole basin from satellites alone. In a region where cyclones meet a low-lying, densely populated coast, intensification lead time becomes evacuation lead time.

**Q. Is this production-ready?**

**A.** No, and we would rather say so. The API is a localhost demonstration service with one worker and no authentication; a whole-field export takes 34 seconds with no caching layer; the data bundle is stale and extending it needs pipeline work; and seventeen ports is not an operational deployment. What is ready is the science, the verification and the interfaces.

**Q. What would you do with three more months?**

**A.** Three things in order. Fix the mixed layer, because the Validation Lab shows that is the one error genuinely available to us to improve. Give the encoder a missing-data channel, because it currently cannot tell a gap from average water. And extend the pipeline to run on live data without a reanalysis target for every day.

## H. Hostile and trap questions

**Q. Your model just memorised the reanalysis, did it not?**

**A.** Through the thermocline it has essentially reached the reanalysis's own accuracy against independent floats, and we say so. But it does that from satellite surface inputs the reanalysis does not use, on a daily cadence, at 24,000 cells. And Feature 17 shows it reproducing a cyclone's cold wake along a moving track - a physical process that appears nowhere in its training objective.

**Q. Blanking 15% of your input makes your model better. Does that not mean your input is useless?**

**A.** No - it means we have a warm bias. Blanked pixels arrive at the network as the channel average, which pulls the prediction cooler, and the best score sits exactly where the bias crosses zero. Past 20% the degradation is real and monotone, reaching +0.50 degC at total blindness with skill going negative at 90%. We rejected the flattering reading of our own curve in writing.

**Q. You have a 14-metre bias in your mixed-layer depth. Why should we trust anything else?**

**A.** Because that is a stage-2 field we explicitly do not ship, and we traced its cause rather than discovering it in a review: +0.19 psu of surface salinity bias against a 0.03 kg/m<sup>3</sup> density threshold - five times the threshold. Our standing internal recommendation is not to use that field. The heat content from the same model, an integral rather than a threshold crossing, has a bias of 0.024 GJ/m<sup>2</sup> and is defensible. Knowing which of your own outputs to distrust is the point.

**Q. There is an error you say you cannot explain. Is that not a serious problem?**

**A.** The Arabian Sea satellite penalty, +0.0341 degC, with the sign holding across all three seeds and the cause unknown after four tested hypotheses. It is 3.8% of our headline error. We report it as unexplained because the alternative is offering a story we have not verified, and this project has already retracted numbers that were believed and wrong.

**Q. Why should we score you well when you admit you invented nothing?**

**A.** Because the problem statement asks for a working, validated, satellite-input reconstruction system for this basin, and that is what we built and verified end to end. We would rather be the team whose numbers survive scrutiny than the team with the larger claim. Every weakness you might find, we have already named - and the numbers that are left standing are the ones that survived our own attempts to break them.

## 9. If you are asked something you cannot answer

- 1** Say so directly. **'I do not know'** and **'that is not verified'** are complete answers, and they are consistent with everything else in this pack.
- 2** Say what *is* known, and where. Almost every open question here has a nearby measured fact - offer that instead of a guess.
- 3** Say how it would be checked. Naming the experiment that would settle it shows you understand the question even when you cannot answer it.

- 4 Never invent a number.** A jury that catches one fabricated figure will discount every other figure you have given them, including the true ones.
- 5** If a juror corrects you and they are right, agree immediately and move on. Do not defend a position you have just learned is wrong.

**THE THREE EVIDENCE TAGS THIS PROJECT RUNS ON - USE THEM OUT LOUD**

**[VERIFIED]** - we ran the code or inspected the data ourselves; only these are facts. **[INFERRED]** - a reasonable reading, not yet tested, and labelled as such. **[UNKNOWN]** - not verified, and said plainly rather than dressed up. Using these aloud in a viva is unusual, and it is immediately convincing.

**IN ONE LINE**

*OceanEmbed reconstructs the North Indian Ocean's subsurface temperature daily from satellites at 0.9006 degC against independent floats - and its most valuable output is an honest account of exactly where it is wrong.*

**16 / 0 / 1**

PS REQUIREMENTS: PASS / FAIL / BLOCKED

**10**

SCIENTIFIC FLAWS, PUBLISHED BY US FIRST

**3**

CONTRIBUTIONS THE PRIOR ART DOES NOT COVER

**Note 19 of 19. End of the jury pack.**